From Data to Decisions: Transforming Zillow Home Values into Pricing Intelligence

A Data-Driven Approach to Market Analysis and Rate Optimization

Actuarial Analysis Project by Tolulope Atanda

The Business Problem Being Solved

Current Pricing Challenges in Homeowners Insurance:

Manual territory reviews take months → Slow response to market changes

Catastrophe risk mispriced in high-growth areas → Adverse selection and loss ratios

Competitors adjust rates faster using alternative data → Market share erosion

Expense loads applied blindly → Missed profit opportunities

Question for Leaders: How quickly can the organization identify mispriced risks and adjust before competitors do?

What Was Built - A Prototype for Real-Time Market Intelligence

- Not just a calculator - a framework for:
- Real-time market monitoring using publicly available Zillow data
- State-specific catastrophe load optimization based on actual risk profiles
- Transparent rate breakdowns that show exactly where premium dollars go
- Key Metric: The model processes 894 metropolitan areas instantly, something that would take a team weeks to do manually.

The Actuarial Math That Drives Business Value

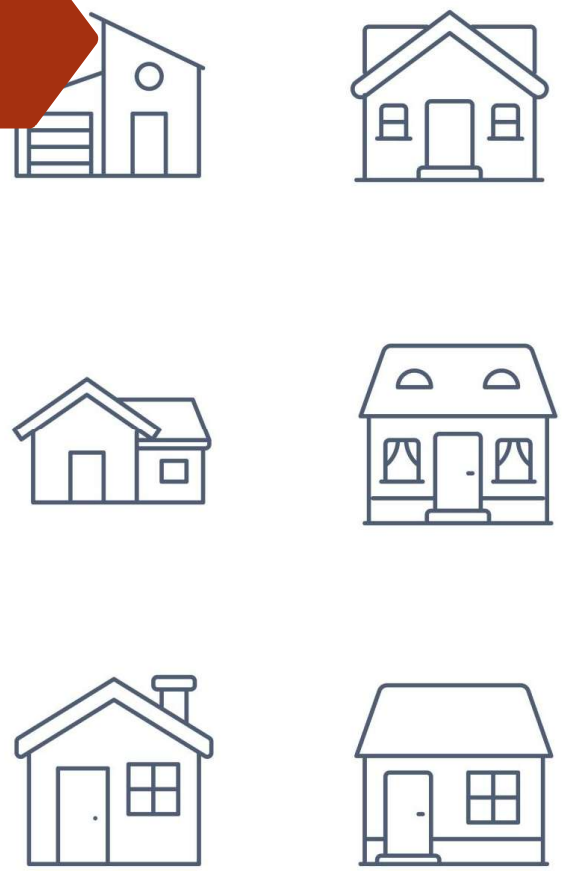
- How each component impacts the bottom line:
- **Replacement Cost (85% factor)**: Excludes land value to avoid over-insuring

  → Prevents moral hazard and reduces claim severity
- **Base Rate by State**: Reflects local loss history and regulation

  → Foundation for rate filings and competitive positioning
- **Catastrophe Load (8-45%):** Prices for hurricane, wildfire, hail risk

  → Directly impacts combined ratio in high-risk zones
- **30% Expense/Profit**: Covers operations and target profit

  → Determines breakeven and ROI thresholds
- **Insight**: Small adjustments to catastrophe loads create a 20-point swing in rate adequacy



Real Data Enables Real Business Decisions

- **What the Zillow data reveals about portfolio risk:**
  - **San Jose, CA:** $1,636,393 home value → $9,493 estimated premium → Wildfire risk
  - **Key West, FL:** $953,369 home value → $8,401 estimated premium → Hurricane risk
  - **Los Angeles, CA:** ~$950,000 home value → ~$7,800 estimated premium → Wildfire/Earthquake
  - **Miami, FL:** ~$850,000 home value → ~$7,500 estimated premium → Hurricane risk
  - **New York, NY:** ~$700,000 home value → ~$6,200 estimated premium → Coastal Storm risk
- **Strategic Implications:**
  - High-value California homes require sophisticated wildfire modeling
  - Florida homes present more volatility despite lower property values
  - Opportunity exists to expand in moderate-risk states across the Midwest and Northeast

Why This Matters - Three Strategic Levers

Lever 1: Competitive Intelligence

A. Compare filed rates to model outputs

B. Identify overpriced or underpriced markets

Lever 2: Catastrophe Risk Management

A. Florida's 45% load vs California's 25% load - adequacy questions remain

B. Model can test profitability scenarios with adjusted loads

Lever 3: Operational Efficiency

A. Automate territory reviews (currently 3-4 months)

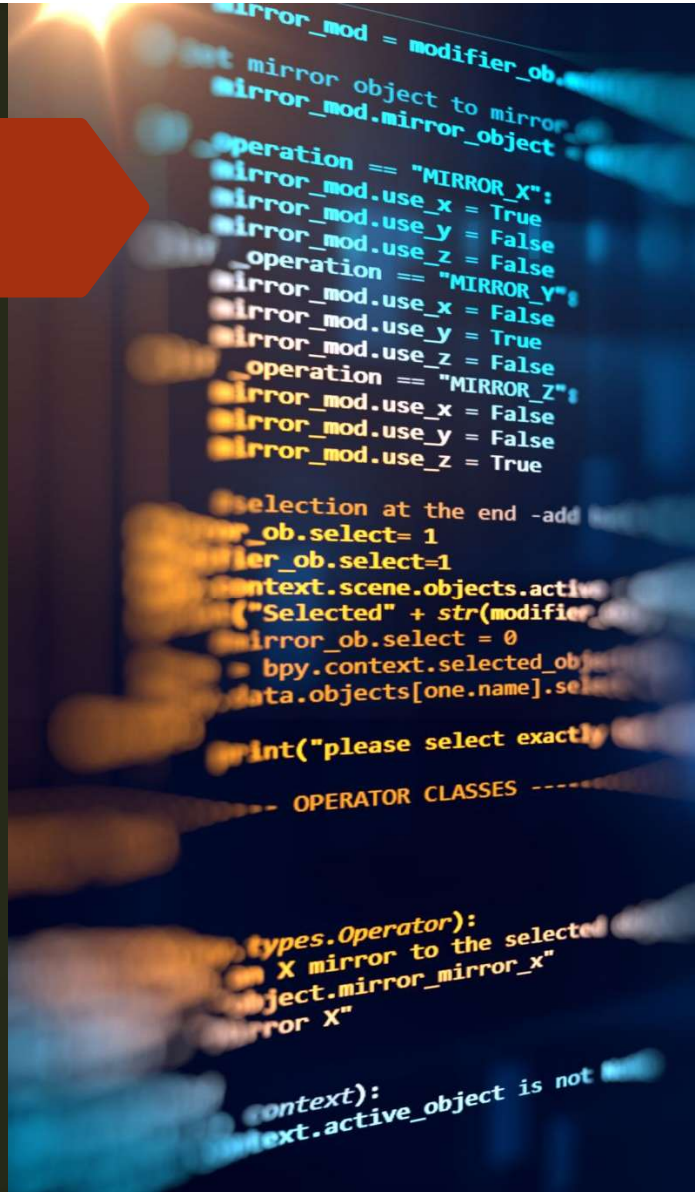
B. Reduce reliance on outdated ISO data

What Senior Leaders Can Do With This Information

- Business Objective → Model Capability → Time to Value
- Identify underpriced territories → Compare loss ratios to model → 2 weeks
- Test rate changes before filing → Adjust cat loads and rerun → 1 day
- Monitor competitor pricing → Track Zillow trends vs book of business → Monthly
- Prepare for investor questions → Show rate adequacy analysis → On demand
- Concrete Example: If Florida loss ratios deteriorate, leadership can request a scenario increasing the cat load from 45% to 55% to see impact on premium and projected loss ratio within minutes.

Next Steps - From Prototype to Production

- **Phase 1 (0-30 days): Validation**
  - Compare model outputs to actual booked premiums for 100 sample policies
  - Calibrate state base rates using internal loss cost data
- **Phase 2 (30-90 days): Integration**
  - Connect to Zillow API for automatic monthly updates
  - Build dashboard for underwriters to see risk scores at point of quote
- **Phase 3 (90+ days): Expansion**
  - Add roof age, construction type, and claims history factors
  - Develop territory-specific rate indications for filing support
- **Resource Estimate**: Partial FTE for data engineering plus actuarial oversight = $50,000-$75,000 to productionize



Questions for the Leadership Team

- For the Actuarial Pricing Team:
  - How close are current base rates to Zillow-driven indications?
  - Which states show the biggest gap between model and actual loss costs?
- For the Product Team:
  - Could the organization offer fast-track quotes in high-value California markets using Zillow data?
  - What is the competitive landscape across the 894 cities that can now be monitored?
- For the Risk and Underwriting Team:
  - How does the catastrophe load compare to the existing reinsurance program?
  - Is the organization inadvertently pricing for wildfire in California without modeling it explicitly?

Closing - The Strategic Opportunity

- **What Competitors Are Doing:**
  - Progressive uses usage-based data
  - Lemonade uses AI and public records
  - Insurtechs use Zillow APIs for instant quoting
- **The Organization's Opportunity:**
  - Catch up quickly with this transparent, explainable model
  - Differentiate by tying rates to actual home values, not just ZIP codes
  - Build actuarial credibility with data-driven rate filings
- **Requested Action:** A 30-day pilot to compare model outputs to the organization's actual book of business in California and Florida